RIVER VALLEY HIGH SCHOOL
JC 2 PRELIMINARY EXAMINATION

CANDIDATE NAME													
CENTRE NUMBER	S					CLASS		INDEX NUMBER					

H2 BIOLOGY **9744/04**

Paper 4 Practical **17 Aug 2021**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number, class and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

Answer **all** questions.

1 In this experiment, you will investigate the effect of substrate concentration on the rate of respiration. Liver tissue was removed from the organism and allowed to respire overnight. Intact mitochondria were then extracted from the liver tissue and placed in a cold buffer solution, labelled **V**.

You are required to:

- make initial observations to determine how to measure the rate of respiration
- perform serial dilution to obtain different concentrations of substrate **S**
- collect results and present it in a suitable table

You are provided with:

- 150 cm^3 of 1.0 % substrate, in a beaker labelled **S** **(HCl)**
- 150 cm^3 of mitochondria suspension, in a beaker labelled **V** **($NaHCO_3$)**
- 200 cm^3 of distilled water, in a beaker labelled **W**
- 5 cm^3 of DCPIP solution, in a container labelled **DCPIP**

DCPIP is an irritant. Suitable eye protection should be worn. If DCPIP comes into contact with your skin, wash it off immediately under tap water.

*Before starting the investigation, read through steps 1-13 and prepare a table in **(c)**.*

Proceed as follows.

1 Using the scissors, cut at least 10 squares of filter paper of dimension 1 cm by 1 cm.

2 Prepare mixture **D** by mixing suspension **V** and solution **DCPIP** in a ratio of 15 to 1. Prepare at least 20 cm^3 of mixture **D** for step 3.

3 Pour mixture **D** into the petri dish to fill it to a height of approximately 2 mm.

4 Place the filter papers into the petri dish. Make sure that the filter papers are completely covered by mixture **D**.

5 You are required to make a serial dilution of 1.0 % substrate **S**, to reduce its concentration by **half** between each successive dilutions.

After the serial dilution is completed, you will need to have at least 40 cm^3 of each concentration available for use.

(a) Complete Fig. 1.1 to show how you will dilute **S**. [3]

For each beaker:

- state, under the beaker, the volume and concentration of the substrate **S** in the beaker that will be available for use in the investigation, after the serial dilution has been completed
- use one arrow, with a label above the beaker, to show the volume and concentration of **S** added to prepare the solution in the beaker
- use another arrow, with a label above the beaker, to show the volume of distilled water, **W**, added to prepare the concentration of **S** in the beaker.

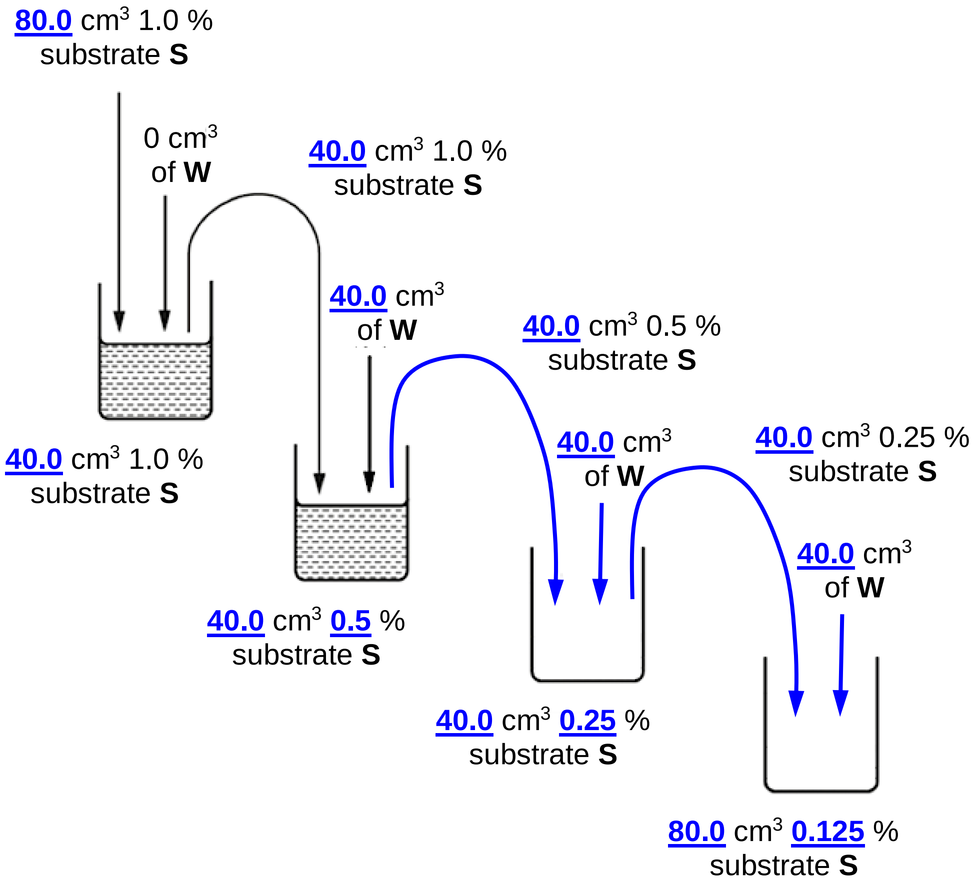

Fig. 1.1

1. **C**: correct concentrations diluted;;
   *1.0%, 0.5%, 0.25%, 0.125% substrate **S** solutions;*

2. **Ca**: correct calculations for all concentrations of **substrate S** and **W**;;
   *Transfer 40.0 cm^3 of substrate **S** and 40.0 cm^3 of **W**;;*

3. **P**: correct precision for volume transferred;
   1 d.p. ;

4. **V**: correct volume of **substrate S** prepared for each concentration;
   At least 40 cm^3 each of concentration ;

6 Using a marker, draw a line to indicate on the specimen tube the position 3 cm from the top of the tube.

7 Pour substrate **S** of concentration 1.0 % into the specimen tube up to this mark.

8 Use the forceps to pick up one square of filter paper from the petri dish. Wipe the square against the petri dish to remove excess solution from both sides of the square.

9 Place the square at the bottom of the specimen tube. Immediately release the square and observe it for one minute.

(b) Describe your observations on the appearance and behaviour of the square of filter paper. [2]

1. Bubbles appeared on the (both) surfaces of the filter paper;
2. Filter paper rose to surface of solution;
3. DCPIP decolourised / blue to light pink;
4. Ref to area on filter paper e.g. completely / partially / half;

MMO – 2

10 Remove the square from the specimen tube and empty the contents of the specimen tube into the container labelled **for waste**.

11 Using your observations in **(b)**, you are required to decide on what you will measure to determine the rate of respiration. You should choose a dependent variable that allows you to observe a trend in the concentration range of substrate **S** prepared.

Describe the dependent variable.

____

12 Use the method you have decided to collect results for all concentrations of **S**. If the time taken is more than 1 minute, record the results as 'more than 60.0'.

13 Calculate the rate of respiration and record in the table in **(c)**.

(c) Record your results in a suitable table in the space below.

Table showing the effect of substrate **S** concentration on the rate of respiration

Concentration of substrate **S** / %	Time taken for filter paper to float to surface, **T** / s or Time taken for filter paper to completely decolourise, **T** / s	Rate of respiration / s^{-1}
1.0		
0.5		
0.25		
0.125		

1. **T**: Title
*Table showing the effect of substrate **S** on the rate of respiration*

2. **H&U**: Heading with units;;
   *Concentration of substrate **S** / % <u>and</u> Time taken for filter paper to completely decolourise / s <u>OR</u> Time taken for filter paper to float to surface / s <u>and</u> Rate of respiration / s^{-1}*

   A: Time taken / s – if student described in full in step 11.

3. **P**: Precision;;
   1 d.p. for time taken; __ s.f. for rate of respiration (depending on student value);

4. **R**: Rate;;
   Correct calculation

5. **Tr**: Trend;;
   Correct trend – increasing time taken / rate

PDO - 4

[4]

(d) **(i)** Complete Table 1.1 to:

- identify one significant source of error for your method
- suggest how to make an improvement to increase the accuracy of your results in **(c)**. [2]

Table 1.1

Method: Measuring time taken for filter paper to float to surface

significant source of error	how to make an improvement
Visual determination of when filter paper reaches surface is subjective;;	Use light sensor attached to data logger to detect when filter paper touches the bottom and reaches the surface;;
Distance travelled by filter paper to reach surface may be different;; OR Path of movement of filter paper floating to surface is different;;	To fill up tube with substrate **S** till line measured from bottom of specimen tube;; Use narrower tube or container so that filter paper floats up directly;;
Temperature of reaction mixtures is not kept the same for different experiments;;	Use thermostatically-controlled water bath at 37 °C to equilibrate and maintain temperatures for all reaction mixtures;;

Method: Measuring time taken for filter paper to completely decolourise

significant source of error	how to make an improvement
Temperature of reaction mixtures is not kept the same for different experiments;;	Use thermostatically-controlled water bath at 37 °C to equilibrate and maintain temperatures for all reaction mixtures;;

Max. one matched pair

ACE - 2

(ii) Suggest the identity of substrate, **S**.

Give a reason for your answer. [2]

1. Pyruvate;;
2. Presence of protein transporters on mitochondrial membrane / uptake of pyruvate into matrix for oxidative phosphorylation;;

ACE - 2

(e) Another experiment was conducted to investigate the rate of respiration in different species of organisms, **X**, **Y** and **Z**. Intact cells were obtained from the same tissue of the different organisms and placed separately in cold buffer solutions.

In the experiment, all other conditions, including the amount of oxygen (in excess), the amount of glucose and the number of cells, were kept the same. 20 minutes after the start of the experiment, a small volume of reaction mixture was removed and Benedict's test was conducted to measure the time taken for first colour change.

(i) Explain how the rate of respiration can be shown by the time taken for first colour change. [2]

1. More glucose taken up by the cell per unit time;
2. Higher rate of glycolysis;
3. More glucose converted (to pyruvate) per unit time;
4. (Less glucose in reaction mixture) resulting in longer time taken for first change in colour for Benedict's test;

ACE – 2

Further investigation was made using the same experimental setup to determine the effect of the amount of protein present in the inner mitochondrial membrane on the time taken for first colour change in the Benedict's test. Table 1.2 shows this relationship.

Using spectroscopic techniques, absorbance measured at 280 nm (A_{280}) was used to determine the amount of protein present in the inner mitochondrial membrane.

The higher the absorbance value, the more proteins present.

Table. 1.2

A_{280} / a.u.	time taken for first colour change / s
1.25	280
1.00	202
0.75	126
0.50	78
0.25	65

(ii) On the grid provided, plot a graph of the data shown in Table 1.2. [3]

1. **H**: Heading with units;;
   x-axis: A_{280} / a.u.;
   y-axis: time taken for first colour change / s;

2. **P:** Precision;
   x-axis: 2 d.p..
   y-axis: whole no.

3. **S**: Scale;
   Graph occupies at least 2/3 of grid AND equal intervals marked on both axes;

4. **PP**: Plot points;
   All points plotted correctly;

5. **L**: line;
   Best fit line;

PDO – 3

(iii) Using **(e)(ii)**, explain what is meant by "limiting factor". [2]

Rate of respiration is limited by

1. Factor that is nearest its minimum value;
2. Which directly affects the rate if its quantity is changed;

Evidence

3. As absorbance at 280nm increases from 0.25 a.u. to 1.00 a.u., time taken for first colour change decreases from 280 s to 78 s;;

ACE – 2

(iv) In the same investigation, during anaerobic conditions, it was found that when A_{280} was 0.75 a.u., the time taken for first colour change was 166 s.

Using the information provided and the same absorbance range, draw the predicted graph for anaerobic conditions on the grid in **(e)(ii)**. [1]

(v) Explain the shape of the graph for **(e)(iv)** and the difference in time taken for first colour change between aerobic and anaerobic conditions when A_{280} was 0.75 a.u.. [2]

Reason for shape of graph:

1. Anaerobic respiration involves enzymes in glycolysis and fermentation;
2. Which are present in the cytosol of cell;

Reason for difference:

3. Shorter time taken due to higher rate of glycolysis during anaerobic conditions;
4. So as to produce similar amount / sufficient amount of ATP;

ACE – 2

The results for the experiment on **X**, **Y** and **Z** are shown in Table 1.3.

Table 1.3

organism	time taken for first colour change / s
X	135
Y	224
Z	92

(vi) The absorbance at 280 nm (A_{280}) can be used to estimate the amount of protein in a mixture using the following formula:

$$\text{concentration (mg / ml)} = A_{280} \times \text{path length of spectrometer (cm)}$$

Given that the path length is 0.8 cm and there are 250 000 cells in 1.5 ml of the reaction mixture tested, use the graph and calculate the amount of protein per cell for the organism with the highest rate of respiration.

Show **all** working clearly.

From graph, A_{280} for organism Y at 224 s = __ ;;

Amount of protein per cell = (__ x 0.8 cm x 1.5 ml) / 250 000

= ________ mg

correct value & precision; units;

PDO – 2

[2]

[Total: 25]

2 Concentrated biological washing liquids are alkali solutions that contain enzymes and other chemicals to help remove stains caused by biological molecules. One of these enzymes is lipase. A student found a website that lists the concentrations of lipase in the concentrated biological washing liquids. These concentrations vary between 85 mg cm^{-3} and 155 mg cm^{-3}.

These concentrated biological washing liquids need to be diluted with water to reach the working concentration of enzyme lipase.

The recommended dilutions for washing fabrics using these concentrated biological washing liquids are:

- Washing liquid **A** – 20 cm^3 of washing liquid in 5000 cm^3 of water
- Washing liquid **B** – 35 cm^3 of washing liquid in 10 000 cm^3 of water.

The student was asked to find the working concentration of enzyme lipase for biological washing liquids, **A** and **B** when diluted.

(a) The student used the information from the website and the recommended dilutions to calculate the working concentration of lipase that could be present in diluted biological washing liquids **A** and **B**.

The student first calculated the **maximum** possible working concentration of lipase in washing liquid **A** when diluted. The method is shown below.

The maximum possible concentration of lipase in concentrated washing liquid **A** is 155 mg cm^{-3}. Therefore, the maximum possible working concentration of lipase in diluted washing liquid **A** is:

$$\frac{(155\ \text{mg cm}^{-3}\ \text{x}\ 20\ \text{cm}^{3})}{5000\ \text{cm}^{3}} = 0.62\ \text{mg cm}^{-3}$$

Calculate the **minimum** possible concentration of lipase in diluted washing liquid **B**. Show all working in the space provided. [2]

$$\left\{\begin{array}{l} 85\ (\text{mg cm}^{-3}) \\ \dfrac{85\ (\text{mg cm}^{-3}) \times 35\ (\text{cm}^{3})}{10\,000\ (\text{cm}^{3})} \end{array}\right\}$$

$= 0.30\ (\text{mg cm}^{-3})$;

Working;;

Correct answer and precision;;

minimum possible concentration of lipase in diluted washing liquid **B:** ________________

To find out the exact working concentrations of lipase in diluted washing liquid **A** and **B**, the student made five different concentrations of lipase, 0.2%, 0.4%, 0.6%, 0.8% and 1.0% and simulated fat-stained fabric by soaking them in cooking oil.

He diluted the concentrated washing liquids **A** and **B** according to the recommended dilutions. The website also recommends that the washes using the diluted washing liquids should be conducted for 1 hour at 30 °C, instead of the optimum temperature of lipase, which is 60 °C.

As the reaction between lipase and cooking oil on the fat-stained fabric proceeds, the pH of the mixture will decrease because of the fatty acids produced from the hydrolysis reaction.

(b) Explain how a lower temperature of 30°C affects the results of the experiment. [2]

1. Lower temperature decreases the rate of lipase activity;
2. Decrease kinetic energy of lipase and fats;
3. Reduce frequency of effective collision between lipase and fats;
4. Hence less enzyme substrate complex formed per unit time;
5. The rate is reduced by 2^3 times (because every 10°C temperature doubles);

(c) Suggest why the recommended temperature to use for washing is 30 °C instead of lipase's optimum temperature of 60 °C. [1]

At higher temperature, longer duration results in more enzymes denaturing before washing cycle is completed (reducing the efficacy of the washing liquid);;

(d) Using apparatus commonly found in a laboratory and the materials below, outline a method the student could use to find the working concentration of lipase in the diluted washing liquid **A** and **B** with high accuracy and reliability.

Data table and risk assessments are **not** needed.

- 0.2%, 0.4%, 0.6%, 0.8% and 1.0% lipase solutions
- diluted washing liquid **A**
- diluted washing liquid **B**
- 10 cm x 10 cm fat-stained fabric
- strings
- boiling tubes
- scissors
- 1% sodium hydroxide
- bromothymol blue which is blue at alkali pH and turns yellow at pH 6.0
- stopwatch [8]

1. Add fixed volume of NaOH to lipase solution at start of experiment;
2. Add fixed volume of bromothymol blue to reaction mixture;
3. Prepare water bath / thermostatically controlled water bath to equilibrate enzyme at 30 °C before adding cotton fabric;
4. Cut fabric to a fixed dimension;
5. Immerse the cloth and start stop watch simultaneously/OWTTE - compulsory point;;
6. Use a stopwatch to record time taken for time taken for reaction mixture to turn yellow (DV marks);;
7. Repeat steps for all lipase concentration and washing liquid A and B;
8. Plot a graph/standard curve using time taken/rate for lipase concentrations 0.2%, 0.4%, 0.6%, 0.8% and 1.0% ;
9. Using the time taken to reach end point for diluted washing liquid A and B and the graph drawn in 8, find working concentration of lipase;

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10. description of 2 controlled variable (e.g. 1 cm3 of NaOH, 2 drops of bromothymol blue; 1 cm x 1 cm fabric);

11. Negative control using boiled enzyme with alkali/water with alkali, expecting that the washing liquid remains blue;;

12. Method to improve accuracy and reliability (replicates and repeat, statistical test);;

(e) A student conducted the experiment and his results suggested that diluted washing liquid **A** had a higher working concentration of lipase than diluted washing liquid **B**.

Suggest one reason why this conclusion may **not** be valid. [1]

Lipase in washing liquid A may have a higher activity that that of B/ Lipases may be extracted from different sources leading to different activity;;

Washing liquid A and B may contain other chemicals not found in the prepared lipase concentration;;

[Total: 14]

3 During this question you will require access to a microscope.

You are provided with:

- a piece of eggplant fruit, **B1**
- iodine solution, in a container labelled **iodine**

Iodine is a stain. If iodine comes into contact with your skin, wash it off with cold water.

You are required to:

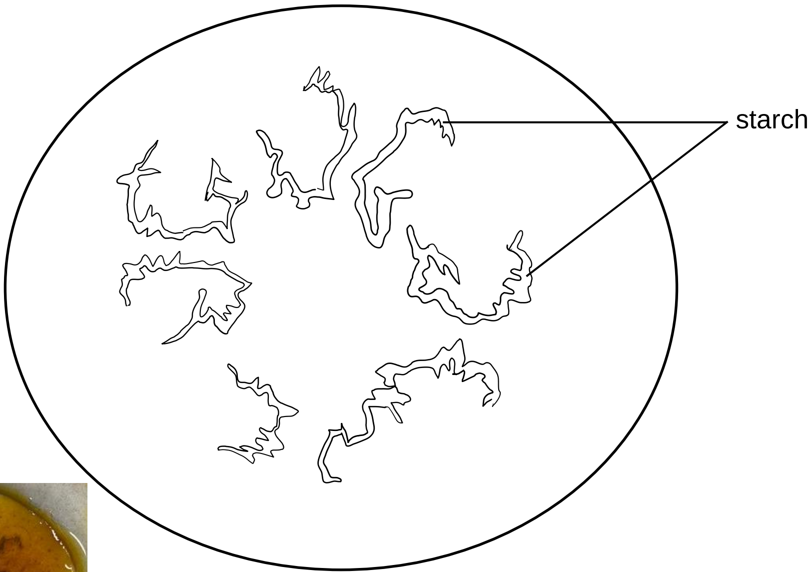
- record observations following the addition of **iodine**
- make a microscope slide of this sample
- draw cells in **B1** observed at high magnification
- calculate the actual size of one cell in **B1**.

Proceed as follows.

1. Cut two transverse slices of eggplant, **B1**. Each slice should be approximately 3 mm thick. Do not remove the skin.
2. Put 5 to 8 drops of **iodine** into one Petri dish.
3. Put one slice of eggplant into the Petri dish. Add more drops of **iodine** to cover the top surface of the slice.
4. Cover the Petri dish with its lid and leave the slice of eggplant for one minute.
5. Observe the slice of eggplant stained with **iodine** through the lid of the Petri dish.

(a) The diagram provided in the space below shows the outline of the slice of eggplant. On the diagram, draw the stained regions observed in **step 5**.

Add labelling lines and the label 'starch' to identify the regions that has starch present. [1]



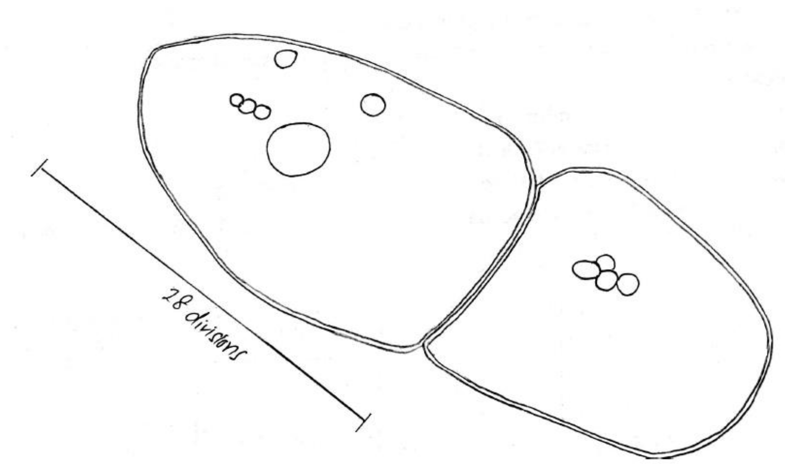
Correct identification of regions with label 'starch';;

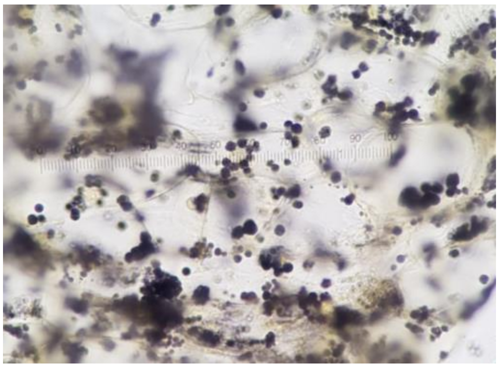
6. Use the forceps to pull out a thin piece of the eggplant from one of the stained regions.
7. Put a drop of water on a glass slide and place the piece of eggplant on the glass slide.
8. Carefully place the cover slip over the piece of eggplant. Place a paper towel on top of the glass slide and press firmly on the cover slip.
9. Use your microscope, at high magnification, view the slide to find two whole cells.

(b) In the space provided, draw **two** adjacent whole cells that you have selected.

You should include in your drawings any internal structures that you can see.

Labels are **not** required. [3]





C: Cell shape;
2 adjacent cells with correct shape

P: Proportion;;
Correct proportion of cell wall
Cell wall of consistent thickness

S: Structure;;
Starch granules with
correct shape
correct size

L: Line;
Continuous lines (i.e. without gaps)
Smooth line of consistent thickness

[penalised ½ m if drawing size < 2/3 space]

(c) Using the stage micrometer and the eyepiece graticule fitted in the microscope, calculate the actual size one cell.

Show **all** working clearly. [3]

1. Calibration of eyepiece graticule at x40 using a stage micrometer

   100 divisions on eyepiece graticule = 25 divisions on stage micrometer
   Each division on stage micrometer = 0.01 mm = 10 μm
   Each division on eyepiece graticule will be:
   (25 division on stage micrometer x 10 μm) / 100 divisions on eyepiece graticule
   = (20 x 10 μm)/ 100;
   = 2.5 μm; correct answer with correct units

2. Number of division overlapping with one cell;;

   R: <5 divisions

3. Show working with correct answer and units;;

   Actual size = number of division overlapping with one cell x 2.5 μm

In the tomato fruit, one method to determine its ripeness is to measure its pH. Tomatoes become less acidic as they ripen. For a perfectly ripe tomato, the pH of its pulp extract is approximately 4.6.

Indicator **G**, can be used to determine the pH. Indicator **G** is orange-red at acidic pH, green at neutral pH and purple-blue at alkaline pH.

P1, **P2** and **P3** are purified tomato pulp extracts from different tomatoes of the same tomato variety but harvested at different times.

You are required to:

- record observations for **P1**, **P2** and **P3** when tested with indicator **G**
- determine and rank the tomato pulp extracts according to the level of ripeness.

Proceed as follows:

1. Add one drop of **G** to one drop of **P1**.

2. Record observations in Table 3.1.

3. Repeat steps 1-2 using **P2** and **P3**.

4. Rank the tomato pulp extract based on its ripeness where 1 indicates unripe and 3 being the most ripe. [2]

(d) **Table 3.1**

	P1	**P2**	**P3**
observations	green colour observed	red / orange-red colour observed	purple colour observed
ripeness	2	1	3

Correct observation for P1, P2 and P3;;

Correct ranking;;

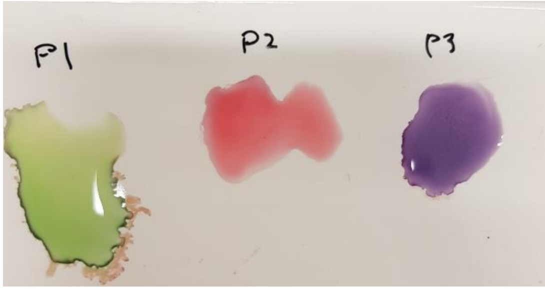


Fig. 3.1 is a photomicrograph of the stem cross-section of a tomato plant while Fig. 3.2 is a photomicrograph of the stem cross-section of another species of plant.

You are not expected to be familiar with this specimen.

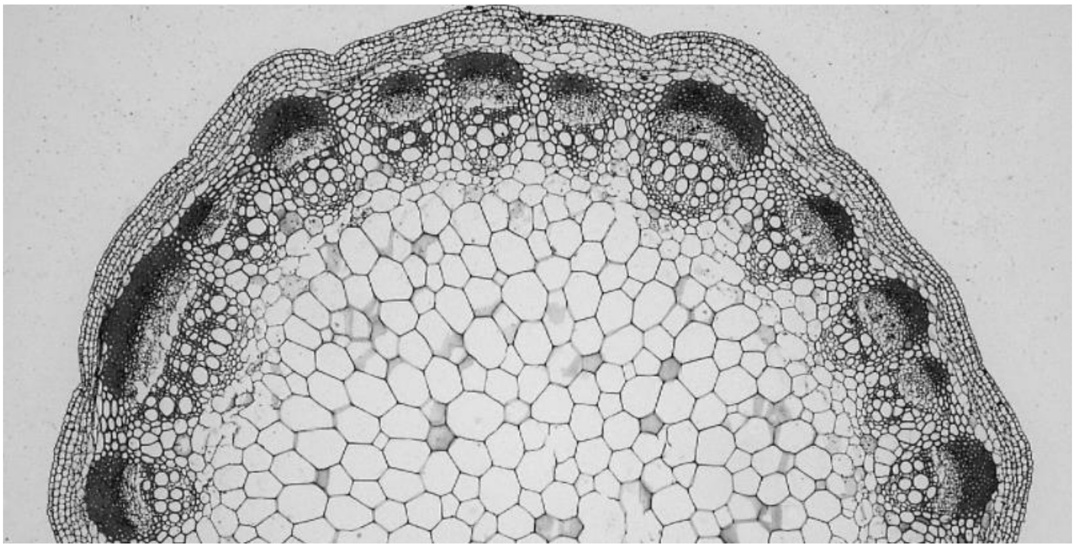


Fig. 3.1

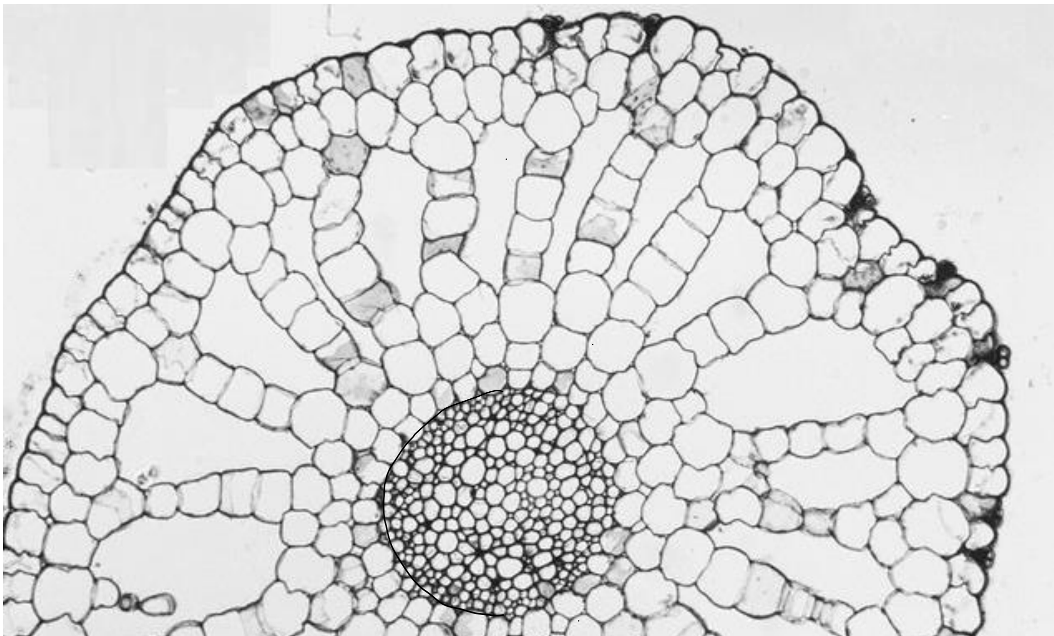
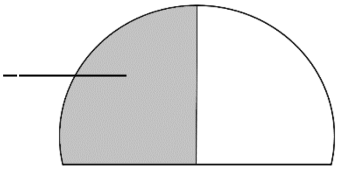


Fig. 3.2

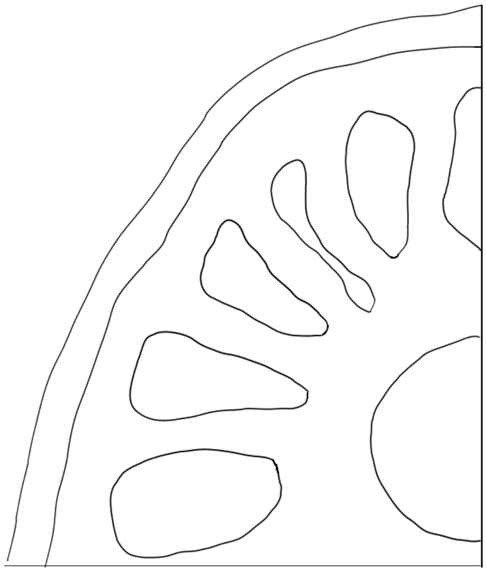
(e) Use the space provided to draw a large plan diagram of the part of the stem shown in Fig. 3.2.



A plan diagram shows the arrangement of different tissues, including their correct shape and proportions. No cells should be drawn.

Labels are **not** required. [4]

1. **P**: Proportion;;
   Epidermis to length;
   vascular bundle to length;
2. **T**: Tissues;;
   3 tissue layer;
   5 air spaces;
3. **S**: Shape;;
   Shape of air spaces;
   Shape of vascular bundle;
4. **L**: Line;;



(f) Using a suitable form, distinguish between the stem structures shown in Fig 3.1 and Fig 3.2. [3]

	Fig 3.2	Fig 3.3
presence of air spaces	absent	present
vascular bundle position	peripheral	central
arrangement of vascular bundle	arranged in a ring	not arranged in a ring

[Total: 16]